Inter-region sparsity experiments

Al Xin

Nov 4, 2026

1 Last seen on ADA...

1.1 Sparsity measures

In neuroscience, a common lifetime sparsity measure is the ratio of moments (call this the "RM" estimator) (Rolls and Tovee, 1995; Vinje and Gallant, 2000):

$$S_{\text{RM}} = \left(1 - \frac{\left(\frac{1}{n}\sum_{i=1}^{n} r_i\right)^2}{\frac{1}{n}\sum_{i=1}^{n} r_i^2}\right)\left(\frac{1}{1-\frac{1}{n}}\right).$$

The term $\left(\frac{1}{1-\frac{1}{n}}\right)$ scales the above to $[0, 1]$, with 0 indicating uniformity and 1 indicating maximal concentration.

The above has some relationship with the pq-mean sparsity measure described by Hurley and Rickard (2009),

$$S_{pq} = -\frac{\left(\frac{1}{n}\sum_{i=1}^{n} r_i^p\right)^{1/p}}{\left(\frac{1}{n}\sum_{i=1}^{n} r_i^q\right)^{1/q}}.$$

The pq-mean sparsity fulfils the "cloning" principle of a sparsity measure, i.e., that the measure, when evaluated on a duplicate of the original population, will produce the same sparsity. Unfortunately, previously mentioned S_{RM} is not invariant to cloning, which makes it unsuitable for between region population sparsity comparisons. It is still usable for per-site comparisons (since the number of presented images is the same for all sites). As another consideration, the pq-mean is not as conveniently bounded by $[0, 1]$.

As another concern, most sparsity measures cannot be calculated when negative values exist in the data. As discussed previously, it might be useful to consider deviation away from baseline when calculating sparsity rather than deviation from zero. Thus, one might want to apply the Hoyer sparsity measure, defined as

$$S_{\text{Hoyer}} = \frac{1}{\sqrt{n}-1}\left(\sqrt{n} - \frac{\sum_{i=1}^{n}|r_i|}{\sqrt{\sum_{i=1}^{n} x_i^2}}\right),$$

which, besides cloning, fills the remaining 5 of Hurley and Rickard's listed desiderata. Furthermore, it is bounded by $[0, 1]$, with 0 indicating complete uniformity and 1 indicating a single non-zero value.

2 Below baseline values

After subtracting the baseline firing, the centered firings show prominent between region differences, with reliable IT sites tending to show more below baseline firing rates (Figure 1).

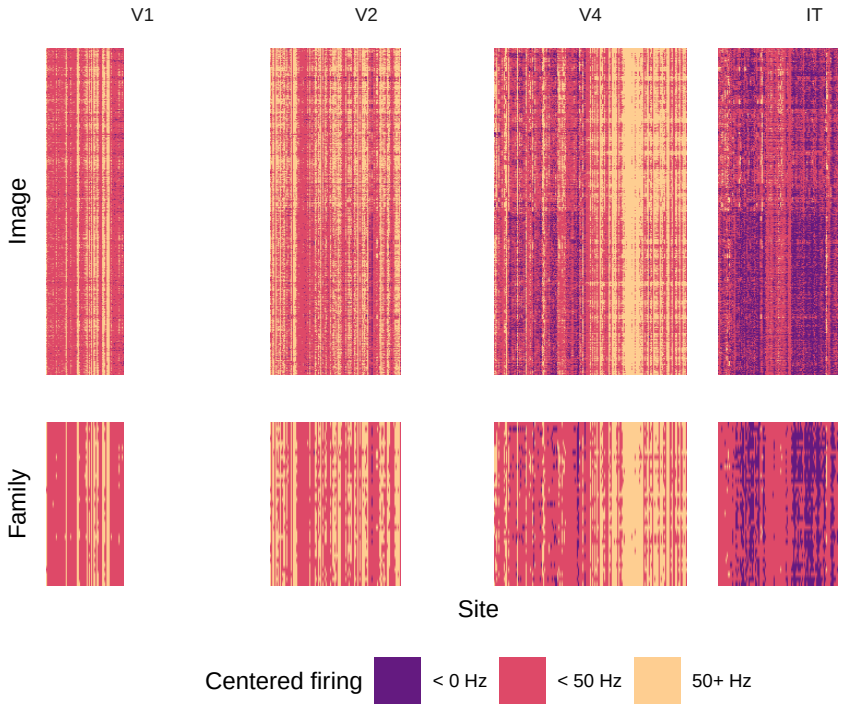


Figure 1: Heat maps of centered/baseline subtracted firing rates, as (top) averaged over specific image repetitions or (bottom) averaged over families. IT shows more sites with below baseline firing. Only sites with split-half correlations > 0.2 are show. See the appendix for all sites.

2.1 Separate sparsities for positive and negative firings

3 Sparsities on log ratios

4 Appendix

4.1 Heat maps

4.2 Different sparsity measures

4.3 Monkey details

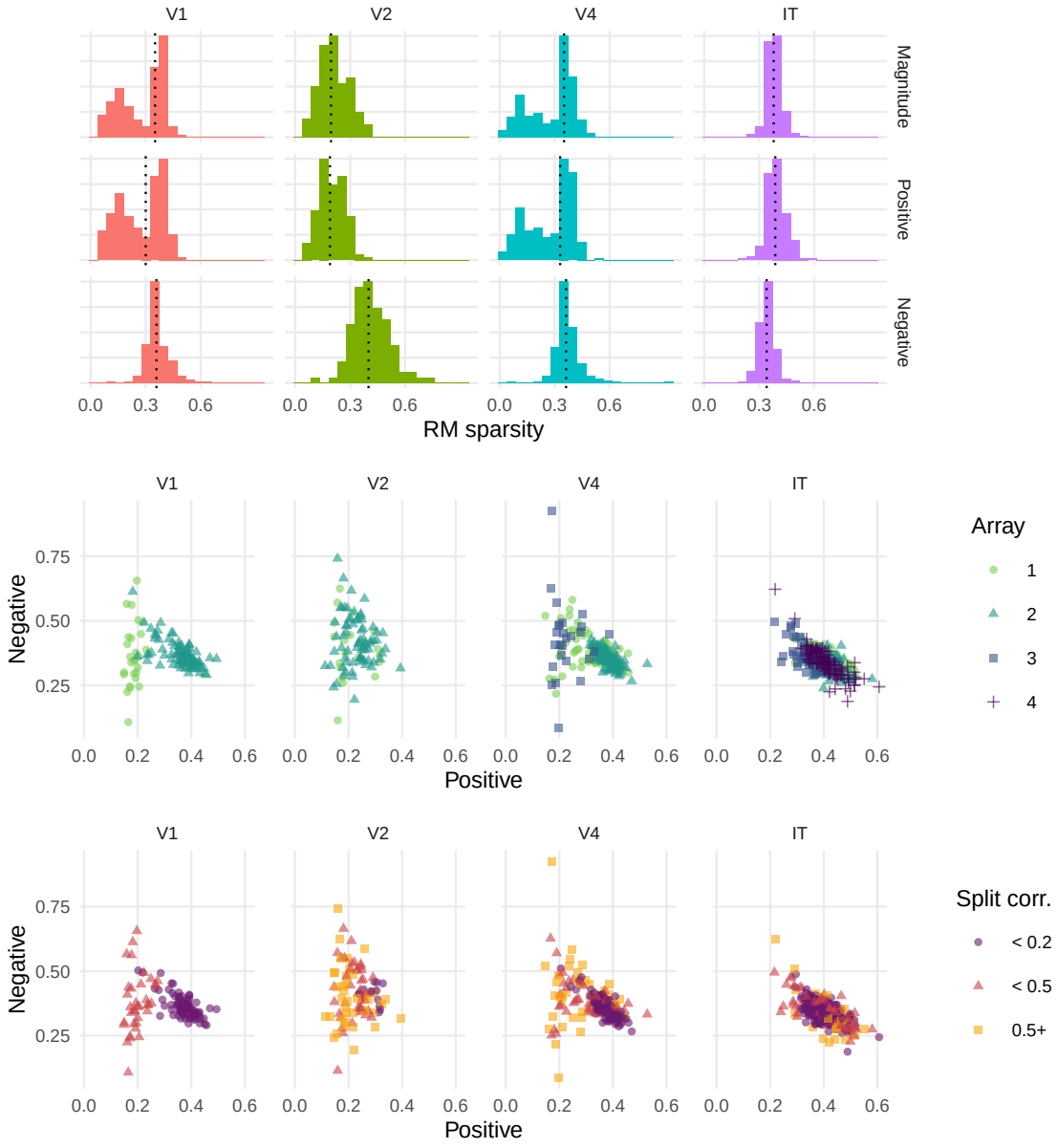


Figure 2: (Top) Histograms of site sparsities, calculated on the absolute value of centered firing (magnitude), only non-negative centered firing (positive), or only negative centered firing values (negative). (Middle) For V1, V4, and IT, each array shows distinct clusters of negative firing and positive firing sparsity patterns. (Bottom) Labeling sites by split-half correlation shows the same clustering.

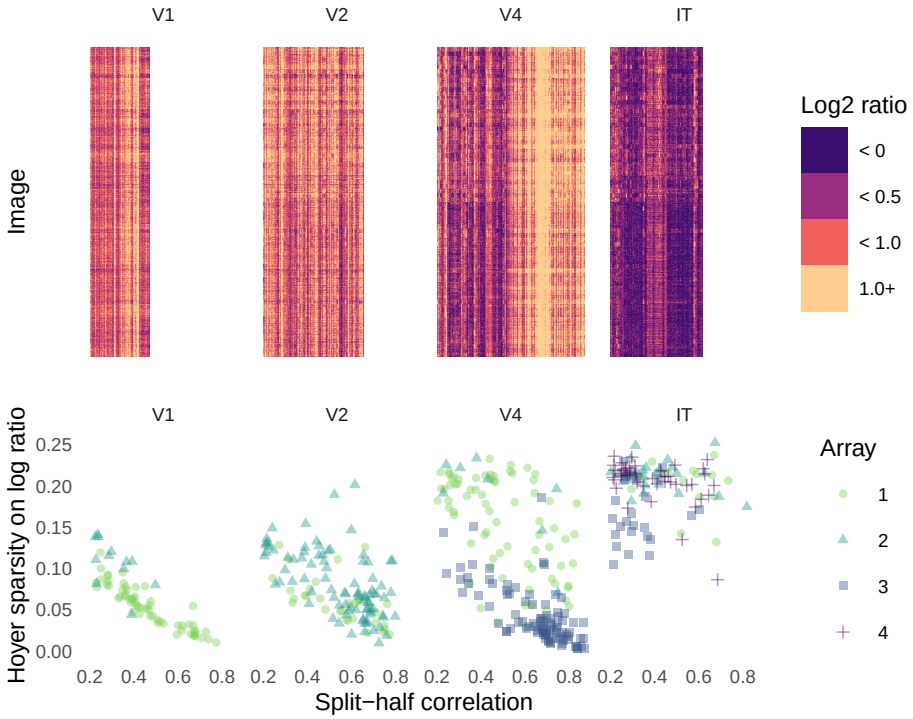


Figure 3: (Top) Heat maps of log (firing / baseline), as averaged over specific image. IT shows more sites with below baseline firing. Only sites with split-half correlations > 0.2 are show. (Bottom) Scatter plots of Hoyer sparsity vs. reliability. Only sites with split-half correlations > 0.2 are show.

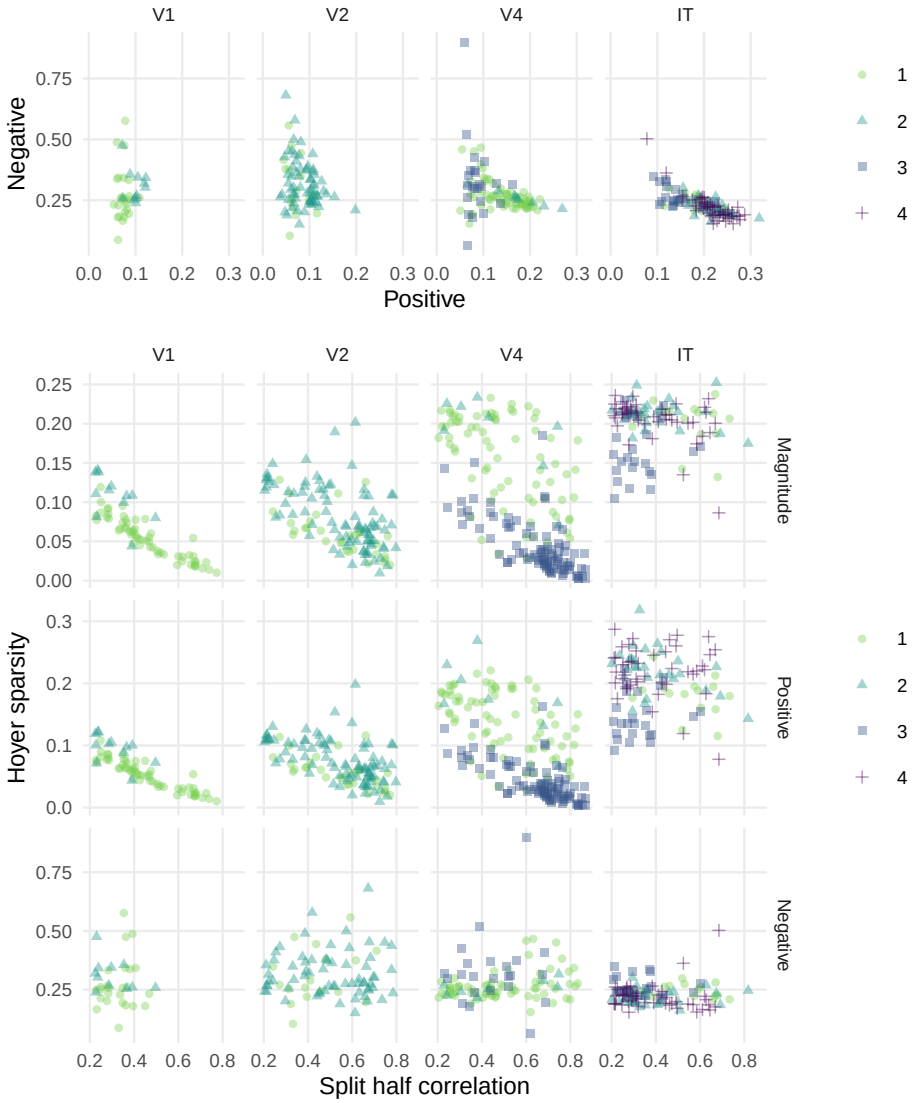


Figure 4: (Top) Scatter plots comparing sparsities calculated on only negative vs. only positive log ratios. (Bottom) Sparsity vs. correlation plots for each category of log ratios (only negatives, only positives, all magnitudes)

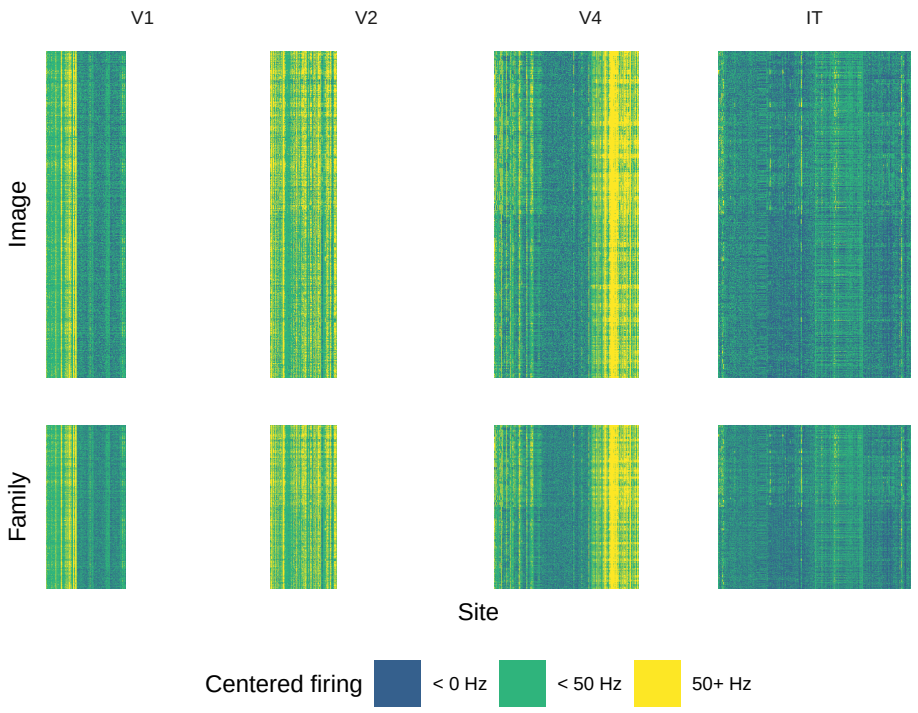


Figure 5: Heat maps of centered/baseline subtracted firing rates, as (top) averaged over specific image repetitions or (bottom) averaged over families. IT shows more sites with below baseline firing.

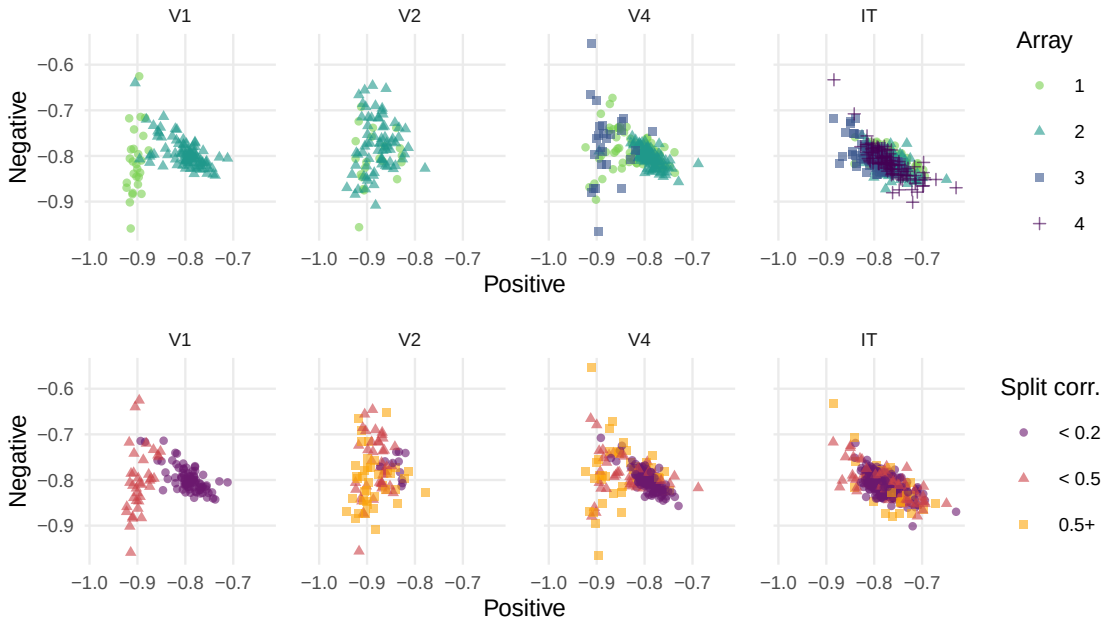


Figure 6

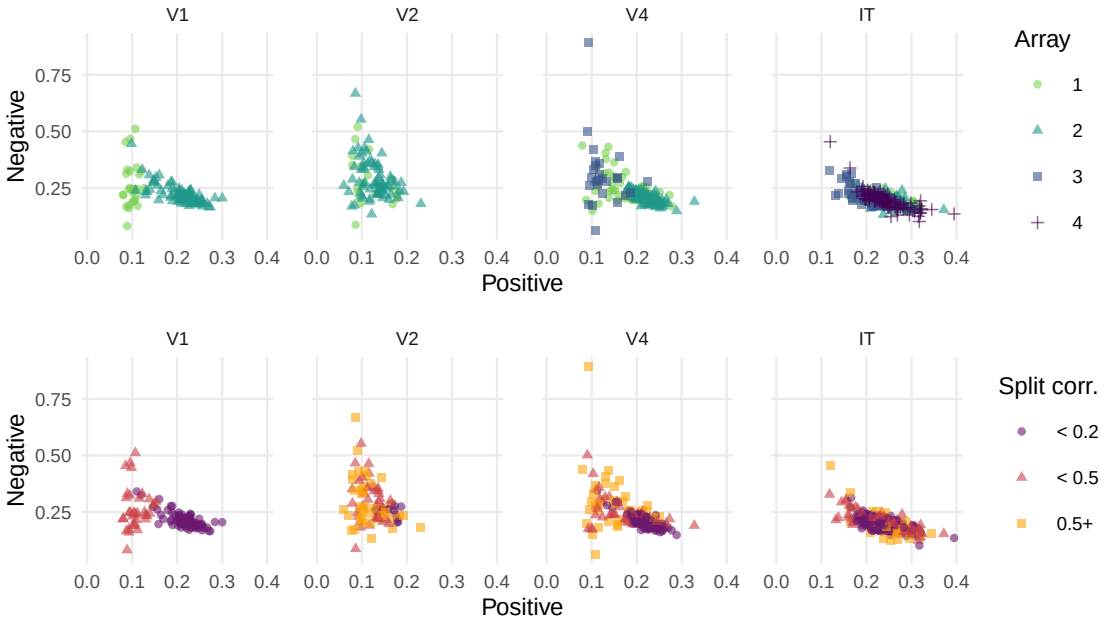


Figure 7

Table 1: File name codes

Region	Array 1	Array 2	Array 3	Array 4
V1	KY (29)	HT (36)		
V2	KY (29)	LI (29)		
V4	KY (29)	LI (29)	XT (409)	
IT	BB (20)	HT (34)	HU (34)	HT (221)

Table 2: Codes corresponding to Table 1 in Lee et al.

Region	Array 1	Array 2	Array 3	Array 4
V1	M1 (29)	M6 (36)		
V2	M1 (29)	M2 (29)		
V4	M1 (29)	M2 (29)	M8 (409)	
IT	M3 (20)	M6 (34)	M7 (34)	M6 (221)

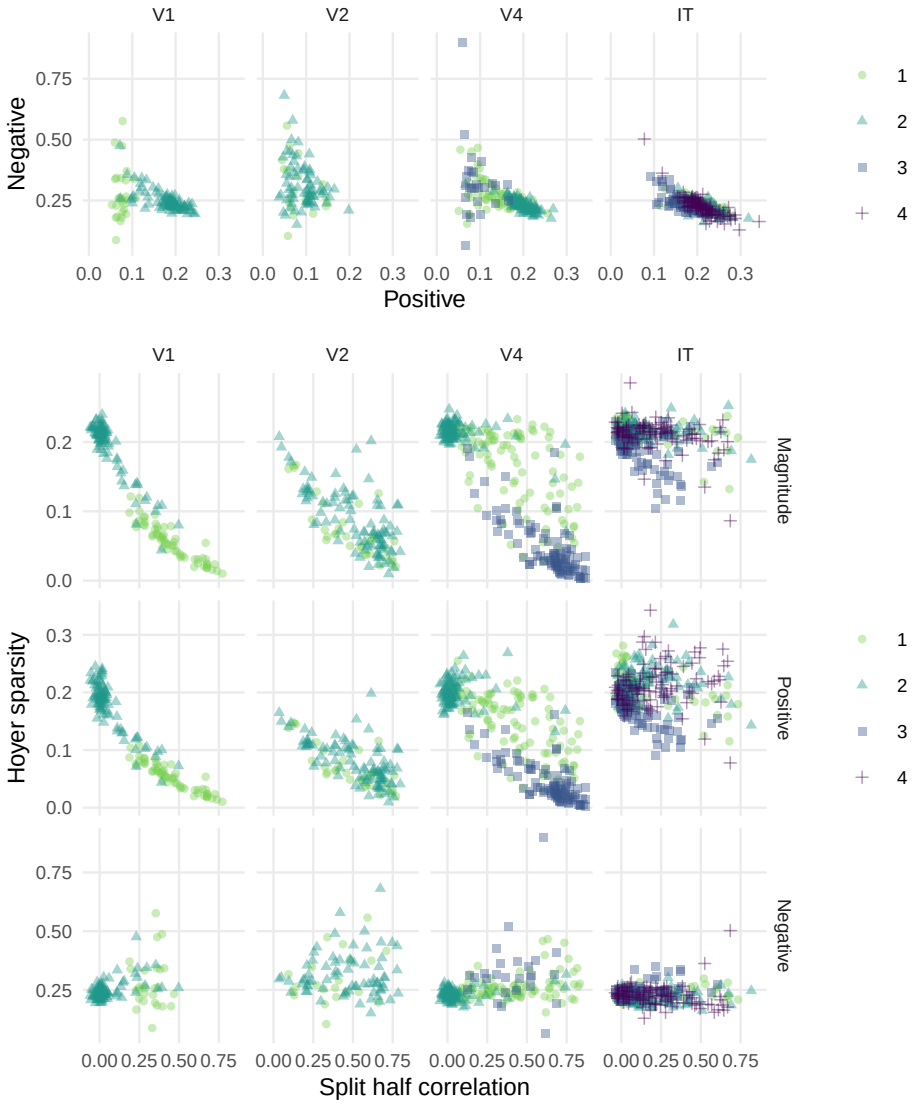


Figure 8: Same as previous figure but without correlation cutoff.

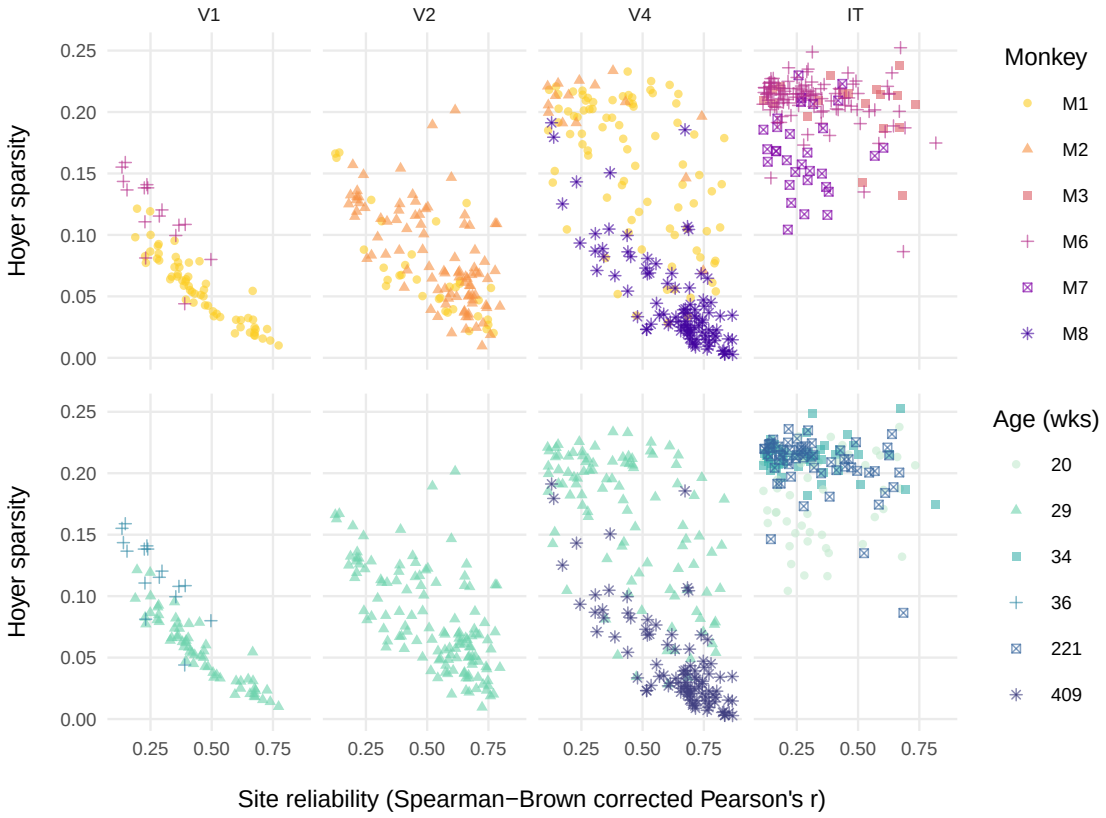


Figure 9: Labeling by monkey rather than array.